

Mobile Sales Dashboard

Project Documentation

Detail	Value
Project Name	Mobile Sales Dashboard
Tool Used	Microsoft Power BI Desktop
Author	Souvik Maity
Learning Source	Power BI course by Satish Dhawale (Skill Course)
Type	Personal learning project

1. Overview

1.1 Objective

To build a single-page, interactive Power BI dashboard that analyzes mobile phone sales data across brands, cities, payment methods, and time, while practicing:

- Data cleaning and transformation in Power Query
- DAX measure creation
- Custom visual design and branding
- Interactive filtering with slicers

1.2 Business Questions Answered

- What are the total sales, quantity sold, and number of transactions overall?
- How does sales volume change month to month and day to day?
- Which cities generate the most revenue?
- Which brands and models sell the most?
- How do customers prefer to pay (UPI, card, cash)?
- How satisfied are customers, based on ratings?

2. Dataset

Detail	Value
Source file	Day - 30 - Mobile Sales Data.xlsx
Sheet used	Sheet 1
Format	Excel workbook (.xlsx)
Granularity	One row per transaction

2.1 Key Columns (post-cleaning)

Column	Description
Transaction ID	Unique identifier per sale
Date	Transaction date (cleaned/derived — see Section 3)
Day Name	Day of week (regenerated — see Section 3)
Brand	Phone manufacturer (Apple, Samsung, OnePlus, Vivo, Xiaomi, etc.)

Mobile Model	Specific phone model
City	Location of sale
Units Sold	Quantity sold in the transaction
Price per Unit	Unit price
Payment Method	UPI / Debit Card / Cash / Credit Card
Customer Ratings	Customer-given rating for the transaction

3. Data Preparation (Power Query)

All cleaning was done in Power Query Editor before loading data into the report.

Step-by-step process:

1. Loaded the source file — Day - 30 - Mobile Sales Data.xlsx, Sheet 1.
2. Changed data types of Day, Month, and Year columns to Text, so they could be concatenated into a single string.
3. Created a custom column named Date using the formula:

`= [Day] & "-" & [Month] & "-" & [Year]`
4. Converted the new Date column's data type from Text to Date.
5. Repositioned the Date column to sit immediately after Transaction ID, for readability.
6. Removed the original Day, Month, and Year columns, since they were now redundant after creating Date.
7. Fixed a data-quality issue in the Day Name column — some values were entered inconsistently (e.g., "sat" instead of "Saturday"). Fixed via Replace Values with Match entire cell contents enabled.
8. Removed the Day Name column entirely and regenerated it cleanly using Select Date column → Add Column → Date → Day Name. This guarantees that day names are derived from the actual date rather than relying on manual/raw entries.
9. Repositioned Day Name directly after Date.
10. Closed & Applied the query to load the cleaned data into the data model.

4. Data Model & DAX Measures

The dashboard uses a single sales table (Sales_Data) containing transaction-level records. Four core DAX measures power the KPI cards and visuals.

```
Total Sales =
SUMX(
    Sales_Data,
    Sales_Data[Units Sold] * Sales_Data[Price per Unit]
)
```

Calculates total revenue by multiplying units sold by price per unit for every row, then summing the result.

```
Total Quantity =
SUM(Sales_Data[Units Sold])
```

Simple sum of all units sold across transactions.

```
Total Transactions =
COUNTROWS(Sales_Data)
```

Counts the number of transactions (rows) in the table.

```
Average Price =  
AVERAGE(Sales_Data[Price per Unit])
```

Average selling price per unit across all transactions.

5. Report Design

5.1 Branding & Color Scheme

The dashboard uses a Motorola-inspired color palette for its visual theme. The color was matched by:

- 11. Inserting the Motorola logo image into PowerPoint.
- 12. Drawing a rectangle shape over it.
- 13. Using Shape Format → Shape Fill → Eyedropper to sample the blue from the logo.
- 14. Copying the resulting hex code: #0060AA.
- 15. Applying that hex code to the Power BI report canvas via Format Page → Canvas Background → Color.

5.2 Layout Elements

- KPI Cards (Total Sales, Total Quantity, Total Transactions, Average Price) arranged in a 2×2 grid.
- Custom callout icons positioned with the KPI cards.
- Rounded rectangle backgrounds behind visuals/cards.
- Consistent shadow treatment, with shadows disabled selectively where visuals overlap to avoid doubled shadow effects.
- Month navigation slicer built from the Date hierarchy using the Month level, with a vertical tile-style layout.

5.3 Visual Inventory

Visual Type	Purpose	Configuration
Line Chart	Total Quantity by Month	X-axis: Date (Month → Day hierarchy, drill-up enabled); Y-axis: Total Quantity; Markers: On; Data labels: On
Clustered Column Chart	Top 3 Models by Sales	X-axis: Mobile Model; Y-axis: Total Sales; Filter: Top N = 3, By value: Total Sales
Pie Chart	Transactions by Payment Method	Legend: Payment Method; Values: Total Transactions
Funnel Chart	Customer Ratings distribution	Category: Customer Ratings; Values: Count of Customer Ratings
Area Chart	Total Sales by Day Name	X-axis: Day Name; Y-axis: Total Sales
Map (Bubble)	Total Sales by City	Location: City; Bubble size: Total Sales; Map settings: Show levels off, Category levels on
Table	Brand-wise summary	Columns: Brand, Total Sales, Total Quantity, Total Transactions; alternating row formatting
Slicers (×4)	Interactive filtering	Fields: Mobile Model, Payment Method, Brand, Day Name — styled as dropdowns

5.4 Build Approach

The dashboard was built incrementally by configuring an initial visual and reusing its formatting when creating subsequent visuals.

- Built and formatted the primary line chart.
- Reused the visual structure to create the Top 3 Models chart.
- Reused the formatting for the payment method and customer rating visuals.
- Built the area chart, table, map, and slicers separately.
- Adjusted shadows and layering to maintain a consistent visual layout.

6. Tools & Technologies

Tool	Purpose
Power BI Desktop	Report building, data modeling, visualization
Power Query Editor	Data cleaning and transformation
DAX	Calculated measures
Microsoft PowerPoint	Used only to extract the visual theme color via the eyedropper tool

7. Key Learnings

- How to derive and clean date-related fields from separate Day/Month/Year columns.
- Writing essential DAX measures using SUMX, SUM, COUNTROWS, and AVERAGE.
- Matching a report's visual theme to a brand-inspired identity using exact color extraction.
- Efficiently reusing and copying visuals and switching chart types instead of rebuilding from scratch.
- Building custom slicer layouts using vertical tile-styled buttons.
- Managing visual layering issues such as overlapping shadows and borders in a dense, single-page dashboard layout.

8. Possible Future Improvements

- Add a Publish to Web live link so viewers can interact with the dashboard directly in a browser without opening Power BI Desktop.
- Add tooltips with more granular transaction-level detail on hover.
- Add YoY/MoM growth analysis when sufficient historical data is available.
- Consider bookmarks for different dashboard views, such as Executive Summary and Regional Deep-Dive.

9. Repository Structure

```
mobile-sales-dashboard/  
├── README.md  
├── dashboard/  
│   ├── Mobile_Sales_Dashboard.pbix  
│   └── README.md  
├── data/  
│   ├── Day - 30 - Mobile Sales Data.xlsx  
│   └── README.md  
├── screenshots/  
│   ├── dashboard_overview.png  
│   └── README.md  
└── docs/  
    └── PROJECT_DOCUMENTATION.md
```

10. Documentation Notes

This document is intended to provide the detailed project methodology and implementation notes. The main README.md should remain concise and focus on the project overview, dashboard features, tools, repository structure, preview, dashboard link, learning source, and author.

Before publishing, verify that the filenames shown in the repository structure exactly match the files committed to GitHub and that the DAX column names match the actual Power BI model.